RKG Wallah Academy

Class 10 Physics — Complete Notes

CBSE | Detailed chapter-wise notes with definitions, laws, derivations, tables, solved numericals and important questions

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• Formula Sheet and SI Units

How to use: Pehle chapter padho, phir diagrams NCERT se apni copy mein banao, numericals khud solve karo aur end ke questions ka answer likh kar practise karo. Diagrams (ray diagrams, circuit diagrams) is PDF mein nahi hain — unhe NCERT dekh kar hi banana hai.

Chapter 1: Light — Reflection and Refraction

1.1 Introduction

• Light is a form of energy that makes objects visible. It travels in a straight line (rectilinear propagation) in a uniform medium. Speed of light in vacuum c = 3×10^8 m/s.

• When light falls on an object it can be reflected (bounce back), refracted (bend while entering another medium) or absorbed.

• Shadow formation, eclipses and pin-hole camera prove that light travels in straight lines.

1.2 Reflection of Light

The bouncing back of light when it strikes a smooth, polished surface is called **reflection**.

Terms

• **Incident ray**: the ray that falls on the surface. **Reflected ray**: the ray that bounces back.

• **Normal**: perpendicular to the surface at the point of incidence.

• **Angle of incidence (i)**: angle between incident ray and normal. **Angle of reflection (r)**: angle between reflected ray and normal.

Laws of Reflection

• **First law:** Angle of incidence is equal to angle of reflection (∠i = ∠r).

• **Second law:** The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.

These laws hold for all types of reflecting surfaces — plane or curved.

Image formed by a plane mirror

• Virtual and erect.

• Same size as the object.

• Image is as far behind the mirror as the object is in front.

• Laterally inverted (left appears right and right appears left).

1.3 Spherical Mirrors

A mirror whose reflecting surface is a part of a hollow sphere is a spherical mirror. If the inner (hollow) surface reflects light it is a **concave mirror** (converging mirror). If the outer bulging surface reflects light it is a **convex mirror** (diverging mirror).

Important terms

Term	Meaning
Pole (P)	Centre of the reflecting surface of the mirror.
Centre of curvature (C)	Centre of the sphere of which the mirror is a part. It is not a part of the mirror.
Radius of curvature (R)	Radius of that sphere; distance PC.
Principal axis	Straight line passing through P and C.
Aperture	Diameter of the reflecting surface (the width of the mirror).
Principal focus (F)	Point on the principal axis where rays parallel to the axis meet (concave: real) or appear to come from (convex: virtual) after reflection.

Term	Meaning
Focal length (f)	Distance between P and F.

Relation: R = 2f or f = R / 2 (valid for mirrors of small aperture)

Rules for drawing ray diagrams (any two rays are enough)

• A ray parallel to the principal axis passes (or appears to pass) through the principal focus after reflection.

• A ray passing through the principal focus (concave) becomes parallel to the principal axis after reflection.

• A ray passing through the centre of curvature C returns along the same path because it falls normally on the mirror.

• A ray incident at the pole P is reflected making equal angle with the principal axis (i = r).

Image formation by a concave mirror

Object position	Image position	Size of image	Nature
At infinity	At focus F	Highly diminished, point-sized	Real, inverted
Beyond C	Between F and C	Diminished	Real, inverted
At C	At C	Same size	Real, inverted
Between C and F	Beyond C	Enlarged	Real, inverted
At F	At infinity	Highly enlarged	Real, inverted
Between P and F	Behind the mirror	Enlarged	Virtual, erect

Image formation by a convex mirror

Object position	Image position	Size of image	Nature
At infinity	At F, behind the mirror	Highly diminished, point-sized	Virtual, erect
Between infinity and P	Between P and F, behind the mirror	Diminished	Virtual, erect

Uses

• **Concave mirror:** shaving mirror, make-up mirror, dentist's and ENT doctor's mirror (enlarged image), torch, headlight and search-light reflectors (parallel beam), solar furnaces and solar cookers (concentrate sunlight).

• **Convex mirror:** rear-view mirror in vehicles (erect, diminished image and wider field of view), street-light reflectors.

1.4 Sign Convention for Spherical Mirrors (New Cartesian)

• Object is always placed to the left of the mirror; light falls from the left.

• All distances are measured from the pole P (origin).

• Distances along the direction of incident light (right of pole, +x) are positive; against it (left, −x) are negative.

• Heights above the principal axis (+y) are positive; below (−y) are negative.

Result: object distance u is always negative. Focal length f is negative for concave mirror and positive for convex mirror.

1.5 Mirror Formula and Magnification

1/v + 1/u = 1/f (u = object distance, v = image distance, f = focal length)

Magnification m = h′/h = − v/u

• h′ = height of image, h = height of object.

• m is negative → real and inverted image. m is positive → virtual and erect image.

• |m| > 1 enlarged, |m| < 1 diminished, |m| = 1 same size.

Q. An object 4 cm high is placed 30 cm in front of a concave mirror of focal length 15 cm. Find the position, size and nature of the image.

Sol. u = −30 cm, f = −15 cm, h = 4 cm. 1/v = 1/f − 1/u = −1/15 + 1/30 = −1/30 → **v = −30 cm**. m = −v/u = −(−30)/(−30) = −1, so h′ = m·h = −4 cm. Image is at 30 cm in front of mirror (at C), real, inverted and of the same size.

Q. A convex mirror has focal length 20 cm. An object is placed 20 cm in front of it. Find image distance and magnification.

Sol. u = −20 cm, f = +20 cm. 1/v = 1/20 + 1/20 = 1/10 → **v = +10 cm**. m = −v/u = −10/(−20) = +0.5. Image is 10 cm behind the mirror, virtual, erect and half the size of the object.

1.6 Refraction of Light

The change in direction (bending) of light when it passes obliquely from one transparent medium to another is called **refraction**. It happens because the speed of light differs in different media.

• Light going from a rarer to a denser medium (air → water) bends **towards** the normal.

• Light going from a denser to a rarer medium bends **away** from the normal.

• A ray falling along the normal (i = 0) passes undeviated.

Refraction through a rectangular glass slab

• Light bends towards the normal at the first surface (air → glass) and away from the normal at the second surface (glass → air).

• The emergent ray is **parallel** to the incident ray but is laterally displaced (shifted sideways).

• Lateral displacement increases with thickness of the slab and the angle of incidence.

Laws of refraction (Snell's law)

• The incident ray, refracted ray and normal lie in the same plane.

• The ratio sin i / sin r is constant for a given pair of media and colour of light.

sin i / sin r = constant = n_{21} (refractive index of medium 2 with respect to medium 1)

Refractive index

• It tells how much a medium can bend light. Absolute refractive index n = speed of light in vacuum (c) / speed of light in the medium (v).

• Relative refractive index $n_{21} = n_2/n_1 = v_1/v_2$.

• Optically denser medium: higher refractive index (light travels slower). Optically rarer medium: lower refractive index.

• Optical density is different from mass density (e.g. kerosene is optically denser than water but less dense by mass).

n = c / v (n has no unit)

Medium	Refractive index	Medium	Refractive index
Air	1.0003	Crown glass	1.52
Ice	1.31	Rock salt	1.54
Water	1.33	Carbon disulphide	1.63
Alcohol	1.36	Dense flint glass	1.65
Kerosene	1.44	Ruby	1.71
Fused quartz	1.46	Sapphire	1.77
Turpentine oil	1.47	Diamond	2.42
Benzene	1.50		

• Diamond has the highest refractive index (2.42) and so light travels slowest in it; this is why diamond sparkles.

1.7 Spherical Lenses

A transparent material bounded by two spherical surfaces is a **lens**. A **convex lens** is thicker at the centre (converging). A **concave lens** is thinner at the centre (diverging).

Term	Meaning
Optical centre (O)	Central point of the lens; a ray passing through it goes undeviated.
Principal axis	Line passing through the optical centre and both centres of curvature.
Principal focus (F)	Point where rays parallel to the axis converge (convex) or appear to diverge from (concave). A lens has two foci, F_1 and F_2.
Focal length (f)	Distance between optical centre and principal focus.
Centres of curvature	Centres of the two spheres; 2F is at distance 2f from O.
Aperture	Effective diameter of the circular outline of the lens.

Rules for ray diagrams of lenses

• A ray parallel to the principal axis passes through the focus F_2 (convex) or appears to come from F_1 (concave).

• A ray passing through the focus (convex) emerges parallel to the principal axis.

• A ray passing through the optical centre goes straight without deviation.

Image formation by a convex lens

Object position	Image position	Size of image	Nature
At infinity	At focus F_2	Highly diminished, point-sized	Real, inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real, inverted
At $2F_1$	At $2F_2$	Same size	Real, inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real, inverted
At F_1	At infinity	Highly enlarged	Real, inverted
Between F_1 and O	Same side as the object	Enlarged	Virtual, erect

Image formation by a concave lens

Object position	Image position	Size of image	Nature
At infinity	At focus F_1	Highly diminished, point-sized	Virtual, erect
Between infinity and optical centre O	Between F_1 and O	Diminished	Virtual, erect

A concave lens always forms a virtual, erect and diminished image whatever the object position. Convex lens uses: magnifying glass, camera, microscope, telescope, spectacles for hypermetropia. Concave lens uses: spectacles for myopia, peep-holes in doors.

1.8 Lens Formula, Magnification and Power

Sign convention for lenses is the same as for mirrors with distances measured from the optical centre. Convex lens: f is positive. Concave lens: f is negative.

$1/v - 1/u = 1/f$ (Lens formula)

Magnification $m = h'/h = v/u$

• m positive → virtual, erect image. m negative → real, inverted image.

Power of a lens

The ability of a lens to converge or diverge light rays is its **power**. It is the reciprocal of focal length (in metres).

P = 1/f (f in metres) Unit: dioptre (D). 1 D = power of a lens of focal length 1 m.

• Power of convex lens is positive; of concave lens negative.

• Lenses in contact: $P = P_1 + P_2 + P_3 + \ldots$ (Used by opticians to combine lenses.)

• Short focal length = high power.

Q. An object is placed 15 cm in front of a convex lens of focal length 10 cm. Find image distance and magnification.

Sol. u = −15 cm, f = +10 cm. 1/v = 1/f + 1/u = 1/10 − 1/15 = 1/30 → **v = +30 cm**. m = v/u = 30/(−15) = −2. Image is real, inverted, twice the size, formed 30 cm on the other side.

Q. A concave lens of focal length 20 cm forms an image of an object placed 40 cm from it. Find the image distance.

Sol. u = −40, f = −20. 1/v = 1/f + 1/u = −1/20 − 1/40 = −3/40 → **v = −13.3 cm**; m = v/u = +0.33. Image is virtual, erect and diminished.

Q. Find the power of a convex lens of focal length 50 cm. If it is combined with a concave lens of power −0.5 D, find net power.

Sol. P = 1/0.5 = +2 D. Net power = 2 + (−0.5) = **+1.5 D**; f = 1/1.5 = 0.67 m = 67 cm.

Important Questions — Chapter 1

Q. State the laws of reflection.

Ans. Angle of incidence equals angle of reflection; incident ray, reflected ray and normal lie in one plane.

Q. Why is a convex mirror used as a rear-view mirror?

Ans. It always forms an erect, diminished image and has a wider field of view, so the driver sees more traffic behind.

Q. What is meant by 1 dioptre?

Ans. Power of a lens whose focal length is 1 metre.

Q. Light enters from air to glass with refractive index 1.50. Find speed of light in glass ($c = 3 \times 10^8$ m/s).

Ans. $v = c/n = 3\times10^8/1.5 = 2 \times 10^8$ m/s.

Q. Why does a ray of light bend when it passes from one medium to another?

Ans. Because the speed of light changes in different media.

Q. Name the mirror that can form a real as well as a virtual image.

Ans. Concave mirror (real when object is beyond F; virtual when object is between P and F).

Chapter 2: The Human Eye and the Colourful World

2.1 The Human Eye

The eye is a natural optical instrument — it works like a camera with a convex lens that forms a real, inverted image on the retina (the screen). The brain interprets it as erect.

Part	Function
Cornea	Transparent bulge at front; most refraction (about two-thirds) of light entering the eye occurs here.
Aqueous humour	Watery fluid between cornea and lens; maintains shape and pressure.
Iris	Coloured muscular diaphragm; controls the size of pupil.
Pupil	Variable aperture; regulates the amount of light entering the eye (small in bright light, large in dim light).
Eye lens	Flexible convex lens; fine-tunes focal length to focus on the retina.
Ciliary muscles	Hold the lens and change its curvature (thickness).
Vitreous humour	Jelly-like fluid filling the eyeball behind the lens.
Retina	Light-sensitive screen with rods (dim light, black-and-white) and cones (bright light, colour vision). Converts light to electrical signals.
Optic nerve	Carries signals from retina to the brain.
Blind spot	Point where optic nerve leaves the retina; no sensory cells, no vision.

Power of accommodation

• The ability of the eye lens to adjust its focal length to see both near and distant objects clearly.

• To see nearby objects ciliary muscles contract → lens becomes thicker → focal length decreases. For distant objects muscles relax → lens becomes thinner → focal length increases.

• **Near point (least distance of distinct vision):** minimum distance at which the eye sees clearly without strain = 25 cm for a normal adult.

• **Far point:** farthest point the eye can see clearly = infinity for a normal eye.

• Normal range of vision: 25 cm to infinity.

• **Persistence of vision:** the image stays on the retina for about 1/16 s; hence motion pictures look continuous.

2.2 Defects of Vision and Their Correction

Defect	Problem	Cause	Correction
Myopia (near-sighted ness)	Can see near objects clearly but distant objects blurred. Far point is nearer than infinity.	Eyeball too long, or eye lens too much curvature (focal length too short). Image forms in front of retina.	Concave lens of suitable power.

Defect	Problem	Cause	Correction
Hypermetropia (far-sightedness)	Can see distant objects clearly but near objects blurred. Near point farther than 25 cm.	Eyeball too short, or focal length too long. Image forms behind retina.	Convex lens of suitable power.
Presbyopia	Difficulty in seeing both near and far objects (old age).	Weakening of ciliary muscles and loss of flexibility of lens.	Bifocal lenses (concave upper, convex lower).
Cataract	Cloudy/milky lens, partial or complete loss of vision.	Opacity of eye lens, usually in old age.	Surgery — lens replaced by artificial lens.

Q. A myopic person has far point at 80 cm. Find power of the lens required to see distant objects clearly.

Sol. The lens must form the image of an object at infinity at the far point. u = −∞, v = −80 cm, so f = −80 cm = −0.8 m. P = 1/f = **−1.25 D** (concave lens).

• Two eyes are better than one: wider field of view (about 150° vs 120°), ability to detect faint objects and 3-D (depth) perception.

• Eye donation: after death the cornea of a donor can restore sight to a corneal-blind person. Eyes should be removed within 4–6 hours of death.

• Reading in dim light or watching TV too close strains the eye. Use proper lighting, rest the eyes and eat vitamin-A-rich food.

2.3 Refraction of Light Through a Prism

• A glass prism has two triangular bases and three rectangular lateral surfaces (inclined to each other).

• **Angle of prism (A):** angle between the two refracting faces. **Angle of deviation (D):** angle between the incident ray and the emergent ray.

• Unlike a slab, the emergent ray is not parallel to the incident ray — it bends towards the base of the prism.

2.4 Dispersion of White Light

• The splitting of white light into its seven constituent colours on passing through a prism is called **dispersion**.

• Order of colours (spectrum): **VIBGYOR** — Violet, Indigo, Blue, Green, Yellow, Orange, Red.

• Different colours bend by different angles. Red bends the least (longest wavelength, fastest in glass); violet bends the most (shortest wavelength, slowest in glass).

• **Newton's experiment:** a second identical prism placed inverted recombines the spectrum into white light, proving that white light is made of the seven colours.

• Rainbow is a natural spectrum of the Sun's light formed after rain. Droplets act as tiny prisms.

Formation of a rainbow

• Occurs when the Sun is behind the observer and rain droplets are in front.

• Steps: refraction and dispersion at first surface of droplet → internal reflection inside → refraction again when leaving.

• The observer sees red at the top and violet at the bottom of the arc.

2.5 Atmospheric Refraction

• Refraction of light by the earth's atmosphere (layers of varying optical density).

• **Twinkling of stars:** Starlight passes through layers of changing refractive index; apparent position of the star changes slightly and its brightness flickers.

• **Planets do not twinkle:** They are much closer, appear as extended sources (collection of point sources), so the variations average out.

• **Advance sunrise and delayed sunset:** The Sun is visible about 2 minutes before actual sunrise and 2 minutes after actual sunset due to atmospheric refraction; the Sun appears slightly flattened near the horizon.

2.6 Scattering of Light

• **Tyndall effect:** scattering of light by colloidal particles (like dust, smoke, tiny water droplets) making the path of light visible. Examples: sunlight in a dense forest, light beam in a smoky room.

• The colour of scattered light depends on the size of the scattering particles. Very fine particles scatter mostly blue light; larger particles scatter longer wavelengths; very large particles scatter white light.

• **Blue colour of the sky:** Air molecules are smaller than the wavelength of visible light and scatter blue (short wavelength) light much more than red. Scattering $\propto 1/\lambda^4$. The scattered blue light reaches our eyes.

• **Colour of the Sun at sunrise and sunset:** The Sun is near the horizon, so light has to travel a longer path through the atmosphere; most of the blue and shorter wavelengths are scattered away, and the red light reaches us — so the Sun appears red.

• **Why the sky looks dark to astronauts:** There is no atmosphere in space, so there is no scattering.

• **Clouds look white:** water droplets are large and scatter all colours almost equally.

• **Danger signals are red:** Red light is scattered the least by fog and smoke, so it can be seen from far away.

Important Questions — Chapter 2

Q. What is the function of the iris and the pupil?

Ans. Iris controls the size of the pupil and so the amount of light entering the eye; the pupil is the opening through which light enters.

Q. Why does a normal eye not see objects closer than 25 cm clearly?

Ans. The ciliary muscles cannot make the lens thick enough (focal length cannot be reduced beyond a limit), so the image cannot be focused on the retina.

Q. What happens to the image distance in the eye when we increase the distance of an object?

Ans. Image distance remains the same (always on the retina); the focal length of the lens changes instead.

Q. Why do we get a rainbow only after rain and with the Sun behind us?

Ans. Raindrops are needed for dispersion and the Sun must be behind so that light enters the droplets and returns to our eyes.

Q. Why is the sky blue and the sunset red?

Ans. Blue is scattered the most by air molecules; at sunset light travels farther and blue is scattered away, leaving red.

Q. Why do stars twinkle but planets do not?

Ans. Starlight undergoes varying atmospheric refraction; planets are extended sources so the variations cancel.

Chapter 3: Electricity

3.1 Electric Charge and Current

• Charge (Q) is a property of matter. SI unit: **coulomb (C)**. Charge of an electron = -1.6×10^{-19} C. So 1 C = 6.25×10^{18} electrons.

• Electric current is the rate of flow of charge. Conventional current flows from positive to negative terminal; electrons actually flow from negative to positive.

> **I = Q / t Unit: ampere (A). 1 A = 1 C/s.**

• Smaller units: 1 mA = 10^{-3} A, 1 μA = 10^{-6} A.

• **Ammeter** measures current; connected in **series**. It has very low resistance.

3.2 Electric Potential and Potential Difference

• Potential difference between two points = work done to move a unit positive charge from one point to the other.

• Cell/battery maintains the potential difference needed to drive current.

> **V = W / Q Unit: volt (V). 1 V = 1 J/C.**

1 volt is the potential difference when 1 joule of work is done to move 1 coulomb of charge between two points. A **voltmeter** measures potential difference; connected in **parallel**. It has very high resistance.

3.3 Circuit Diagram

Component	Symbol/meaning
Cell	Long thin line (+) and short thick line (−)
Battery	Combination of cells
Plug key (open / closed)	Switch to complete or break circuit
Resistor	Zig-zag line or rectangle
Variable resistor (rheostat)	Resistor with arrow — changes resistance
Ammeter	Circle with A
Voltmeter	Circle with V
Wire joint / crossing	Dot = joined; no dot = not joined

3.4 Ohm's Law

Statement: At constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends.

> **$V \propto I \rightarrow V = I R$ (R = resistance of the conductor)**

• The V–I graph for an ohmic conductor is a straight line passing through the origin.

• **Resistance:** the property of a conductor to resist the flow of charges through it. SI unit: **ohm (Ω)**. 1 Ω = 1 V/A.

• 1 ohm: resistance of a conductor if a current of 1 A flows through it when the potential difference across its ends is 1 V.

• A variable resistor (rheostat) is used to change current without changing the voltage source.

3.5 Factors on Which Resistance of a Conductor Depends

• Directly proportional to length l.

• Inversely proportional to area of cross-section A.

• Depends on the material and temperature.

$R = \rho l / A$ (ρ = resistivity, unit: Ω m)

• Resistivity is a characteristic property of a material (does not depend on dimensions) but changes with temperature.

• Metals and alloys have low resistivity (good conductors); insulators (glass, rubber) have very high; semiconductors are in between.

• Alloys have higher resistivity than pure metals and do not oxidise (burn) readily at high temperatures → used in heating devices (e.g., nichrome in heaters, toasters, irons).

• Copper and aluminium are used for electric transmission lines because of low resistivity; tungsten is used for bulb filaments because of its high melting point.

Material	Resistivity (Ω m)	Type
Silver	1.60×10^{-8}	Conductor
Copper	1.62×10^{-8}	Conductor
Aluminium	2.63×10^{-8}	Conductor
Tungsten	5.20×10^{-8}	Conductor
Nichrome (alloy)	1.0×10^{-6}	Alloy
Glass	10^{10} – 10^{14}	Insulator
Rubber	10^{13} – 10^{16}	Insulator

3.6 Resistors in Series

• Resistors are joined end to end. The same current flows through each resistor. Total potential difference is the sum of individual drops.

Derivation: $V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 = I(R_1 + R_2 + R_3)$. Also $V = I R_s$. Hence:

$R_s = R_1 + R_2 + R_3 + \ldots$

• Equivalent resistance is greater than the largest individual resistance.

• If one component fails the whole circuit breaks.

3.7 Resistors in Parallel

• Resistors are connected between the same two points. Potential difference across each is the same. Total current is the sum of branch currents.

Derivation: $I = I_1 + I_2 + I_3 = V/R_1 + V/R_2 + V/R_3$. Also $I = V/R_p$. Hence:

$$\mathbf{1/R_p = 1/R_1 + 1/R_2 + 1/R_3 + \ldots}$$

- Equivalent resistance is less than the smallest individual resistance.
- Two resistors: $R_p = R_1R_2/(R_1 + R_2)$.
- **Advantages of parallel connection in homes:** each appliance gets the same voltage (220 V), works independently (if one fails others work), and has its own switch; total resistance is low so a large current can be drawn when needed.

3.8 Heating Effect of Electric Current

When current flows through a resistor, electrical energy is converted to heat. Work done by the source = V I t, so heat produced:

$$\mathbf{H = I^2 R t \quad (Joule's\ law\ of\ heating) \quad (also\ H = V I t = V^2 t / R)}$$

- Heat is directly proportional to the square of the current, resistance and time.
- **Applications:** electric heater, iron, toaster, kettle, electric bulb (filament becomes white hot), fuse.
- **Fuse:** a safety device with a wire of low melting point (alloy of tin and lead) placed in series. When current exceeds a limit, the wire melts and breaks the circuit, protecting the appliance.
- **Bulb filament** is made of tungsten (melting point 3380°C) and the bulb is filled with inert gases like nitrogen and argon to prolong filament life.
- Wires of a heater are not heated like its element because the supply wires have low resistance but the heating element has high resistance (same current flows through both).

3.9 Electric Power and Energy

$$\mathbf{P = W / t = V I = I^2 R = V^2 / R \quad Unit: watt\ (W).\ 1\ W = 1\ V \times 1\ A.}$$

- 1 kilowatt = 1000 W. Energy = power × time.
- SI unit of energy: joule (J). Commercial unit: **kilowatt-hour (kWh)** = 1 unit.

$$\mathbf{1\ kWh = 3.6 \times 10^6\ J}$$

Solved Numericals — Chapter 3

Q. A current of 0.5 A flows through a wire for 10 minutes. How much charge flows?

Sol. Q = I t = 0.5 × 600 = **300 C**.

Q. A copper wire has length 100 m and area of cross-section 1 mm^2. Find its resistance ($\rho = 1.62 \times 10^{-8}\ \Omega$ m).

Sol. A = 1 $mm^2 = 10^{-6}\ m^2$. R = ρ l/A = $1.62\times10^{-8} \times 100 / 10^{-6}$ = **1.62 Ω**.

Q. Three resistors 2 Ω, 3 Ω and 5 Ω are connected in series to a 10 V battery. Find current and voltage across each.

Sol. R = 2 + 3 + 5 = 10 Ω. I = V/R = 1 A. Voltages: 2 V, 3 V and 5 V (sum = 10 V).

Q. 6 Ω and 3 Ω resistors are connected in parallel to a 12 V supply. Find equivalent resistance and branch currents.

Sol. $1/R = 1/6 + 1/3 = 1/2 \rightarrow R = \mathbf{2\ \Omega}$. $I_1 = 12/6 = 2$ A, $I_2 = 12/3 = 4$ A, total I = 6 A.

Q. A bulb is rated 100 W, 220 V. Find its resistance and current.

Sol. $R = V^2/P = 220^2/100 = \mathbf{484\ \Omega}$. $I = P/V = 100/220 \approx \mathbf{0.45\ A}$.

Q. A 4 Ω resistor carries 3 A for 10 s. Find the heat produced.

Sol. $H = I^2Rt = 9 \times 4 \times 10 = \mathbf{360\ J}$.

Q. A 2 kW heater is used 5 hours daily for 30 days. Find the energy consumed and bill at Rs 6 per unit.

Sol. Energy = 2 × 5 × 30 = **300 kWh**. Bill = 300 × 6 = **Rs 1800**.

Important Questions — Chapter 3

Q. Define 1 volt and 1 ampere.

Ans. 1 volt = 1 J of work done per coulomb of charge. 1 ampere = flow of 1 coulomb per second.

Q. Why is tungsten used for filaments of bulbs and not for heater coils?

Ans. Tungsten has a very high melting point; heater coils need a high resistivity alloy that does not oxidise (nichrome).

Q. Why are series arrangements not used in domestic circuits?

Ans. If one device fails the whole circuit breaks; devices cannot be operated independently and they would not get full voltage.

Q. How does the resistance of a wire change if its length is doubled?

Ans. It doubles.

Q. What is the resistance if the radius of the wire is doubled?

Ans. Area becomes 4 times, so resistance becomes 1/4.

Q. Why is copper preferred for connecting wires?

Ans. It has very low resistivity and is ductile, so less heat is lost in the wires.

Chapter 4: Magnetic Effects of Electric Current

4.1 Magnetic Field and Field Lines

• A magnet has two poles: North and South. Like poles repel, unlike poles attract. A freely suspended magnet always rests in the north–south direction.

• **Magnetic field:** the region around a magnet in which its force can be detected. It is a vector quantity with both magnitude and direction.

• **Field lines:** imaginary lines showing direction of field. By convention they emerge from the north pole and merge at the south pole outside the magnet; inside the magnet they go from south to north (closed curves).

Properties of field lines

• Direction of the field at a point is along the tangent to the line.

• Closer lines mean a stronger field; they are closest near the poles.

• Two field lines never intersect (otherwise at the point of intersection there would be two directions).

• Field lines are closed curves.

4.2 Magnetic Field due to a Current-carrying Conductor

• **Oersted's experiment (1820):** A compass needle placed near a current-carrying wire gets deflected, showing that electric current produces a magnetic field around it. Deflection reverses when the current direction reverses.

(a) Straight conductor

• Field lines are concentric circles around the wire in a plane perpendicular to it.

• **Right-hand thumb rule:** Hold the wire in the right hand with the thumb along the direction of current; curled fingers show the direction of the magnetic field.

• Magnetic field strength increases with current and decreases with distance from the wire.

(b) Circular loop

• Field lines near the wire are concentric circles; at the centre they appear as straight lines perpendicular to the plane of loop.

• The field at the centre is stronger when current is more, radius is smaller or the number of turns is greater. For a coil of n turns the field is n times that of a single turn.

(c) Solenoid

• A coil of many circular turns of insulated copper wire wound in the shape of a cylinder.

• The field pattern is similar to a bar magnet: one end behaves as north pole and the other as south pole. Field inside is uniform (parallel straight lines).

• Strength of field depends on the current, the number of turns per unit length and the core material.

• **Electromagnet:** A soft iron core inside the solenoid gets magnetised when current passes and loses magnetism when the current stops. Used in cranes (to lift scrap), electric bells, relays, MRI.

• Permanent magnets are made of steel (retain magnetism); electromagnets of soft iron.

4.3 Force on a Current-carrying Conductor in a Magnetic Field

• Andre-Marie Ampere suggested that a magnet exerts a force on a current-carrying conductor, and by Newton's third law the conductor exerts equal and opposite force on the magnet.

• Force is maximum when the current is perpendicular to the field and zero when it is parallel to it.

• Direction of force reverses if the direction of the current or of the field is reversed.

> **Fleming's Left-Hand Rule: Stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular. If the forefinger points in the direction of the field and the middle finger in the direction of current, the thumb gives the direction of force (motion) on the conductor.**

Memory trick: **F**ore finger = **F**ield, **C**entre (middle) finger = **C**urrent, **T**humb = **T**hrust (force). Used for the electric motor.

4.4 Electric Motor

A device that converts electrical energy into mechanical energy. **Principle:** a current-carrying coil placed in a magnetic field experiences a torque.

Part	Function
Armature coil (ABCD)	Rectangular coil of insulated copper wire that rotates.
Strong field magnet (N–S)	Provides the magnetic field.
Split-ring commutator (C_1, C_2)	Reverses the direction of current every half rotation so the coil keeps rotating in the same direction.
Brushes (B_1, B_2)	Carbon contacts that connect the battery to the split rings.
Axle	Shaft of the rotor connected to the load.

• **Working:** Current flows in the coil; by Fleming's left-hand rule, the two arms AB and CD experience equal and opposite forces and the coil rotates. After half a turn the commutator reverses the current, so forces still push the coil in the same direction → continuous rotation.

• To increase the speed/power: increase current, number of turns, area of the coil or strength of the magnet.

• Commercial motors use an electromagnet and a large number of turns on a soft iron core (armature).

• Uses: fans, mixers, washing machines, pumps, electric vehicles.

4.5 Electromagnetic Induction (EMI)

• **Faraday's discovery:** a changing magnetic field (or relative motion between a coil and a magnet) induces a current in the coil — called induced current; the emf is induced emf.

• Moving a bar magnet into a coil connected to a galvanometer gives deflection; the deflection is opposite when the magnet is taken out; no deflection when the magnet is

stationary.

• Faster motion or more turns gives greater induced current.

> **Fleming's Right-Hand Rule: Stretch the thumb, forefinger and middle finger of the right hand mutually perpendicular. If the forefinger points to the direction of the field and the thumb to the direction of motion of the conductor, the middle finger gives the direction of the induced current.**

4.6 Electric Generator

Converts **mechanical energy into electrical energy** based on electromagnetic induction. **Principle:** when a coil rotates in a magnetic field, the magnetic flux changes and a current is induced.

• **AC generator:** a rotating rectangular coil between the poles of a magnet; its ends are connected to two slip rings (R_1, R_2) that rub against carbon brushes (B_1, B_2). The coil rotates by an external mechanical source (turbine, wind, etc.). Output polarity reverses after every half rotation → alternating current.

• **DC generator:** uses a split-ring commutator instead of slip rings, so the output current flows in one direction only.

• **AC (alternating current):** changes direction periodically. In India, frequency is 50 Hz (changes direction every 1/100 s) and supply voltage is 220 V.

• **DC (direct current):** flows in one direction (e.g., from batteries/cells).

• **Advantage of AC:** it can be transmitted over long distances without much loss of energy. **Disadvantage:** cannot be stored.

4.7 Domestic Electric Circuits

• Power comes to homes through **live wire (red insulation, positive, 220 V)** and **neutral wire (black insulation, 0 V)**. The **earth wire (green insulation)** is connected to a metal plate deep in the earth near the house.

• Earth wire is connected to the metallic body of appliances like refrigerator, iron, geyser; it provides a low-resistance path to ground for leakage current and prevents electric shock.

• Main fuse and a distribution board are used; separate circuits: 5 A for lights and fans (low power) and 15 A for high-power appliances like AC, geyser, refrigerator.

• **Short circuit:** when live and neutral wires touch (damaged insulation) the resistance becomes very low and the current becomes very high, causing heating and fire.

• **Overloading:** too many appliances connected to a single socket draw more current than the wire can carry.

• Safety devices: fuse, MCB (miniature circuit breaker), earthing.

Important Questions — Chapter 4

Q. State the rule that gives direction of magnetic field around a current-carrying wire.

Ans. Right-hand thumb rule (Maxwell's corkscrew rule).

Q. What is the role of the split ring in an electric motor?

Ans. Reverses the direction of current in the coil after each half rotation so that the coil continues to rotate in the same direction.

Q. Name some devices that use electromagnets.

Ans. Electric bell, crane, loudspeaker, MRI machine, relay.

Q. Differentiate between AC and DC generators.

Ans. AC generator has slip rings (current reverses); DC generator has split-ring commutator (current in one direction).

Q. Why is the earth wire used with metallic appliances?

Ans. To discharge any leakage current safely to the ground and prevent electric shock.

Q. What is a short circuit and what causes overloading?

Ans. Live and neutral touching causes a short circuit; connecting many high-power devices to one circuit causes overloading.

Q. When is the force on a current-carrying conductor in a magnetic field maximum?

Ans. When the conductor is perpendicular to the magnetic field.

Chapter 5: Sources of Energy (optional)

Note: This chapter has been removed from the recent CBSE syllabus. Read it only if your school/board still includes it.

5.1 What is a Good Source of Energy?

• Provides large amount of useful energy per unit mass/volume (high calorific value).
• Easy to store and transport.
• Cheap, safe, easy to handle and causes low pollution.
• Gives a steady and uniform supply of energy.

Calorific value = amount of heat produced on complete burning of 1 kg of fuel (kJ/kg). Wood has lower, coal medium, and LPG/hydrogen very high calorific values.

5.2 Conventional Sources of Energy

Fossil fuels

• Coal, petroleum and natural gas formed from the remains of plants and animals buried millions of years ago. They are **non-renewable** and will be exhausted.
• Burning gives CO_2 (global warming), SO_2 and NO_x (acid rain) and smoke.
• To reduce pollution, technologies like using chimney filters and fuel-efficient stoves are used.

Thermal power plant

• Coal/petroleum is burnt → steam → turbine rotates → generator produces electricity.

Located near coal or oil fields because transporting electricity is cheaper than transporting fuel.

Hydroelectric power plant

• Potential energy of stored water in a dam changes to kinetic energy → turbine → electricity. Renewable but needs large dams that submerge land and displace people and destroy ecosystems; also methane is produced from rotting submerged vegetation.

Improvement in technology

• **Biomass:** wood, cow dung, agricultural waste. Charcoal burns without flames, with less smoke and higher heat.
• **Biogas:** produced by anaerobic decomposition of cow dung, sewage and plant waste in a biogas plant. It is mainly methane (75%), burns without smoke, leaves residue (slurry) that is a good manure, and is a clean fuel.
• **Wind energy:** Wind turns turbine blades connected to a generator. A wind farm of many turbines is needed. Needs wind speed above 15 km/h and large land; initial cost is high.

5.3 Alternative or Non-conventional Sources

• **Solar energy:** Sun gives about 1.4 kW/m^2 (solar constant) above the earth's atmosphere. Used in solar cookers (reflector mirror + black-painted box + glass lid) and solar cells (photovoltaic cells made of silicon convert sunlight to electricity). Solar panels power satellites, calculators and street lights. Costly and weather dependent.

• **Tidal energy:** Rise and fall of sea level due to the gravitational pull of the Moon is used to turn turbines.
• **Wave energy:** Kinetic energy of waves near the sea shore is used.
• **Ocean thermal energy (OTEC):** Difference in temperature between surface and deep sea water is used (needs ≥ 20 °C difference).
• **Geothermal energy:** Hot rocks inside the earth heat underground water to steam which turns turbines. Natural hot springs.
• **Nuclear energy:** Released when heavy nuclei are split or light nuclei are fused; mass lost is converted to energy: $E = mc^2$.

Nuclear fission and fusion

Fission	Fusion
Heavy nucleus (e.g. uranium-235) splits into lighter nuclei on bombardment with slow neutrons.	Light nuclei (hydrogen isotopes) combine to form a heavier nucleus at extremely high temperature and pressure.
Used in nuclear reactors and atom bomb.	Source of energy in the Sun and stars; hydrogen bomb.
Produces radioactive waste.	Clean, but difficult to achieve on earth (needs millions of degrees).

• Nuclear power plants: Tarapur (Maharashtra), Rawatbhata (Rajasthan), Kalpakkam (Tamil Nadu), Narora (U.P.), Kakrapar (Gujarat), Kaiga (Karnataka).
• Hazards: nuclear waste disposal, radiation leaks (Chernobyl, Fukushima), and high cost.

5.4 Environmental Consequences and Conservation

• Every source has some environmental cost; there is no perfect source.
• Renewable sources (solar, wind, hydro, biomass) are sustainable; fossil fuels should be used carefully.
• Practise saving: switch off unused lights, use LEDs, public transport and energy-efficient appliances.

Important Questions — Chapter 5

Q. Why should we look for alternative sources of energy?
Ans. Fossil fuels are limited, non-renewable and polluting; demand is increasing.

Q. What are the advantages of biogas?
Ans. Smokeless, high heat value, no solid waste, residue is manure, reduces waste, good for rural areas.

Q. Why can wind energy be harnessed only in certain areas?
Ans. Because strong and steady wind is needed (speed above 15 km/h).

Q. What are the limitations of solar cells?
Ans. Costly, depend on sunlight, special silicon is expensive, and need storage.

Q. Give two differences between nuclear fission and fusion.
Ans. See the table above.

Formula Sheet and SI Units

Topic	Formula
Mirror formula	$1/v + 1/u = 1/f$; $R = 2f$
Mirror magnification	$m = h'/h = -v/u$
Lens formula	$1/v - 1/u = 1/f$
Lens magnification	$m = h'/h = v/u$
Power of lens	$P = 1/f$ (metres), $P_{net} = P_1 + P_2$
Refractive index	$n = c/v$; $n = \sin i / \sin r$
Current	$I = Q/t$
Potential difference	$V = W/Q$
Ohm's law	$V = IR$
Resistance	$R = \rho l/A$
Series	$R_s = R_1 + R_2 + \ldots$
Parallel	$1/R_p = 1/R_1 + 1/R_2 + \ldots$
Heat	$H = I^2Rt = VIt = V^2t/R$
Power	$P = VI = I^2R = V^2/R$
Energy	$E = Pt$; $1\ kWh = 3.6 \times 10^6\ J$
Nuclear energy	$E = mc^2$

Quantity	SI unit	Symbol
Focal length / distances	metre	m
Power of lens	dioptre	D
Charge	coulomb	C
Current	ampere	A
Potential difference	volt	V
Resistance	ohm	Ω
Resistivity	ohm-metre	Ω m
Electric power	watt	W
Energy	joule	J
Frequency	hertz	Hz

Best of luck, topper! ✍ Roz thoda padho, formula yaad karo aur numericals practise karo.